

Unit 3: Biodiversity, Threats & Conservation Strategies

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Levels of Biodiversity: Genetic, species (richness & evenness), and community (α, β, γ \alpha, \beta, \gamma diversity Biogeographic Classification: The 10 distinct biogeographic zones of India Value of Biodiversity: Consumptive, ...

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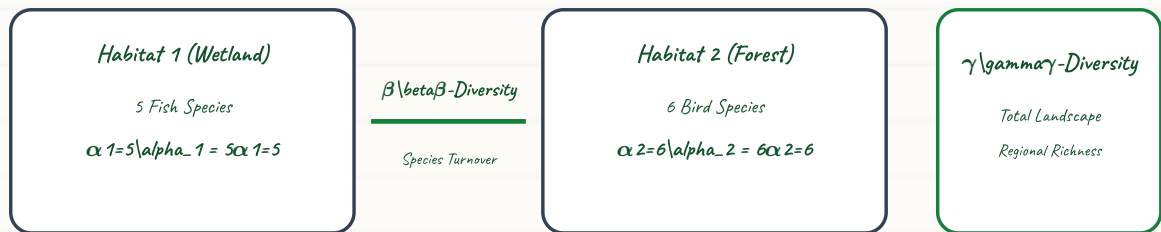
1. Concept & Levels of Biodiversity

Definition:

Biological Diversity (Biodiversity) refers to the structural and functional variety and variability among all living organisms from all sources (terrestrial, marine, and other aquatic ecosystems) and the ecological complexes of which they are part. Coined by Walter G. Rosen in 1985 and popularized by sociobiologist Edward O. Wilson.

1.1 The Three Hierarchical Levels of Biodiversity

1. **1. Genetic Diversity:** The total genetic information contained in the genes of individuals of a single species.
 - High genetic variation enables populations to adapt to evolutionary pressures, resist novel viral pathogens, and survive climatic fluctuations.
 - *Example:* India alone possesses over **50, 000** distinct strains of cultivated rice (*Oryza sativa*) and over **1, 000** varieties of mango (*Mangifera indica*).
2. **2. Species Diversity:** The variety of species within a specific biological community or geographic region. Consists of two distinct ecological dimensions:
 - *Species Richness (S):* The simple raw count of different species present in a community.
 - *Species Evenness (Equitability, J'):* The relative abundance and equitability with which individuals are distributed among the various species.
3. **3. Ecosystem / Community Diversity:** The structural and functional diversity of ecological habitats, biotic communities, and ecological processes across the biosphere. Ecologist R.H. Whittaker partitioned this into three spatial scales:
 - **Alpha (α) Diversity:** Species diversity within a single local, homogeneous habitat or community.
 - **Beta (β) Diversity:** The rate of species turnover (differentiation) along an environmental gradient between adjacent distinct habitats.
 - **Gamma (γ) Diversity:** The overall total species richness across a large regional landscape or entire biome.

Figure 3.1: Spatial Scales of Ecosystem Diversity (α , β , γ)

α measures local richness, β measures compositional change between habitats, and γ measures landscape-wide richness.

Quick Recall:

Three levels: Genetic (allelic variation within species), Species (richness + evenness), and Ecosystem (α = within-habitat, β = between-habitat turnover, γ = landscape total).

2. Biogeographical Classification of India

The Wildlife Institute of India (Rodgers and Panwar, 1988) classified India into **10 Biogeographic Zones** reflecting distinct geographic, climatic, and floral/faunal compositions:

Zone Number & Name	Geographic Coverage	Characteristic Keystone Flora & Fauna
1. Trans-Himalaya	High altitude Tibetan plateau (Ladakh, Lahaul-Spiti).	Snow Leopard, Tibetan Wild Ass (Kiang), Wild Yak, Black-necked Crane.
2. Himalaya	Northwestern, Western, Central, and Eastern Himalayan ranges.	Musk Deer, Himalayan Tahr, Red Panda, Rhododendrons, Deodar.
3. Desert	Arid Thar desert (Rajasthan) and Great Rann of Kutch (Gujarat).	Great Indian Bustard (critically endangered), Desert Fox, Chinkara.
4. Semi-Arid	Transition belt between desert and Deccan (Gujarat, Rajasthan, Punjab).	Asiatic Lion (Gir Forest, Gujarat), Blackbuck, Nilgai, Caracal.
5. Western Ghats	Malabar plains and Sahyadri mountain chain along the west coast.	Lion-tailed Macaque, Nilgiri Tahr, Malabar Grey Hornbill, Evergreen Rainforests.
6. Deccan	Largest zone (43% of India's)	Tiger, Indian Elephant, Sambar, Gaur

Zone Number & Name	Geographic Coverage	Characteristic Keystone Flora & Fauna
7. Gangetic Plain	Alluvial river basin of Ganges and Yamuna rivers.	Gangetic River Dolphin (National Aquatic Animal), Gharial, One-horned Rhino.
8. Coasts	East and West coastlines, Sundarbans mangrove delta.	Olive Ridley Sea Turtles (Gahirmatha beach nesting), Royal Bengal Tiger in Sundarbans.
9. North-East	Assam valley and Cachar hills (high rainfall biodiversity hotspot).	Hoolock Gibbon (India's only ape), Great Indian One-horned Rhinoceros (Kaziranga).
10. Islands	Andaman & Nicobar (Bay of Bengal) and Lakshadweep (Arabian Sea).	Dugong (Sea Cow), Nicobar Megapode, Coral reefs, giant robber crabs.

Quick Recall:

Rodgers & Panwar classified India into 10 zones: Trans-Himalaya, Himalaya, Desert, Semi-Arid, Western Ghats, Deccan Peninsula, Gangetic Plain, Coasts, North-East, and Islands.

3. Values of Biodiversity

Ecological economists evaluate biodiversity across direct and indirect value streams:

- **1. Consumptive Use Value:** Direct harvest of natural products for subsistence without passing through a formal commercial market (e.g. firewood, edible forest fruits, wild medicinal plants consumed directly by forest-dwelling tribes).
- **2. Productive Use Value:** Commercial products harvested, processed, and traded in national/international markets:
 - *Pharmaceuticals:* Over **25%** of prescription medicines originate from plant alkaloids (e.g., **Quinine** from *Cinchona* bark for malaria; **Taxol** from *Taxus* yew tree for chemotherapy; **Vincristine** from Madagascar periwinkle *Catharanthus roseus* for leukemia).
 - *Industrial materials:* Natural rubber, silk, timber, essential oils, tannins, and resins.
- **3. Social & Ethical Value:** Cultural respect and traditional sanctity attached to flora and fauna (e.g. *Ocimum sanctum* [Tulsi], *Ficus religiosa* [Peepal], Sacred Groves preserved untouched across rural India).

- **4. Aesthetic & Ecotourism Value:** Psychological recreation and revenue generated by national parks and wildlife sanctuaries.
- **5. Ecosystem Service Value:** Unpriced fundamental life-support functions: insect pollination of agricultural crops (worth billions of dollars), watershed retention, carbon sequestration, and soil formation.

Quick Recall:

Values of biodiversity: Consumptive (direct domestic use), Productive (market commodities e.g. Quinine, Taxol), Social/Ethical (sacred groves), and Ecosystem Services (pollination, carbon sequestration).

4. India as a Mega-Diversity Nation

Mega-Diversity Status:

India is officially designated as one of the world's **17 Mega-Diversity Nations** by Conservation International. Despite occupying merely **2.4%** of the world's total land area, India contributes an astonishing **7 – 8%** of all recorded biological species on Earth!

- **Floral Wealth:** Over **48,000** plant species (representing **11%** of global flora), of which nearly **33%** are strictly endemic to India.
- **Faunal Wealth:** Over **102,000** recorded animal species (including **3,000+** fish, **500+** amphibians, **1,300+** bird species, and **400+** mammals).
- **Agrobiodiversity:** Center of origin for major crops (rice, sugarcane, pigeonpea, mango, turmeric, black pepper).

Quick Recall:

India holds 17-megadiverse status: **2.4%** of global surface supports **7 – 8%** of global species, including **> 48,000** plant and **> 102,000** animal species.

5. Biodiversity Hotspots (Norman Myers Criteria)

Definition:

A **Biodiversity Hotspot** is a biogeographic region characterized by exceptional levels of plant endemism experiencing severe, irreversible habitat destruction.

5.1 The Two Strict Criteria (Norman Myers, 1988)

To qualify as a global biodiversity hotspot, a region must strictly satisfy two empirical thresholds:

1. **Exceptional Endemism:** It must support at least **1,500** species of endemic vascular plants ($> 0.5\%$ of the world's total vascular flora) found nowhere else on Earth.
2. **High Degree of Threat:** It must have lost at least **70%** of its original primary native habitat (possessing $\leq 30\%$ of its pristine natural vegetation remaining).

5.2 The 4 Biodiversity Hotspots Covering India

Globally, 36 biodiversity hotspots are recognized. Four of these intersect the sovereign territory of India:

- **1. Western Ghats & Sri Lanka:** Continuous mountain chain running parallel to India's west coast through Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu. Exhibits extraordinary endemism in amphibians ($> 75\%$ endemic), reptiles, and flowering plants. Home to the endangered Lion-tailed Macaque and Nilgiri Tahr.
- **2. Eastern Himalayas:** Encompasses Bhutan, northeastern India (Arunachal Pradesh, Sikkim), and northern Myanmar. Rich in alpine rhododendrons, orchids, Red Panda, and Snow Leopard.
- **3. Indo-Burma:** Covers northeastern states south of the Brahmaputra river (Assam, Manipur, Mizoram, Nagaland) extending through Myanmar, Thailand, and Vietnam.
- **4. Sundaland:** Includes the southern Nicobar Islands of India, extending across Indonesia, Malaysia, and Borneo. Characterized by lush mangrove swamps and tropical rainforests.

Quick Recall:

Myers criteria: $\geq 1,500$ endemic vascular plant species AND $\geq 70\%$ primary habitat lost. Four Indian hotspots: Western Ghats, Eastern Himalayas, Indo-Burma, and Sundaland.

6. Threats to Biodiversity (The "Evil Quartet")

E.O. Wilson characterized the major drivers of modern anthropogenic extinction as the *Evil Quartet*:

- **1. Habitat Destruction & Fragmentation:** The leading driver of extinction. Clearing pristine forests for agriculture, urbanization, or highways fractures contiguous biological ranges into isolated ecological fragments, triggering genetic inbreeding and edge-effect degradation.
- **2. Over-Exploitation & Poaching:** Commercial hunting of wildlife for international black markets (e.g. elephant ivory poaching, tiger bones and pelts for pseudo-traditional medicines, rhinoceros horns).

- **3. Invasive Alien Species Introductions:** Non-native species introduced intentionally or accidentally that outcompete native flora and fauna:
 - *Parthenium hysterophorus* (Congress Grass / Carrot Grass): Toxic weed outcompeting native forage.
 - *Lantana camara*: Toxic shrub overtaking Indian forest understories.
 - *Eichhornia crassipes* (Water Hyacinth / "Terror of Bengal"): Clogs water bodies, blocking sunlight and asphyxiating aquatic biodiversity.
- **4. Co-Extinctions & Climate Change:** When a host species becomes extinct (e.g. a specific pollinating fig wasp), obligate dependent symbionts inevitably perish.

Quick Recall:

The Evil Quartet: 1. Habitat loss & fragmentation (primary cause); 2. Over-exploitation (poaching); 3. Invasive alien species (*Lantana*, *Parthenium*, *Eichhornia*); 4. Co-extinction.

7. Endangered & Endemic Species of India

7.1 IUCN Red List Categories

The International Union for Conservation of Nature (IUCN) classifies species into standardized conservation risk tiers:

- **Extinct (EX):** No living individual remains on Earth (e.g. Asiatic Cheetah declared extinct in India in 1952; Dodo).
- **Critically Endangered (CR):** Extremely high risk of extinction in the wild (e.g. Great Indian Bustard, Gharial, Pygmy Hog).
- **Endangered (EN):** Very high risk of extinction in the wild (e.g. Asian Elephant, Tiger, Lion-tailed Macaque, Ganges River Dolphin).
- **Vulnerable (VU):** High risk of endangerment (e.g. One-horned Rhino, Nilgiri Tahr).
- **Near Threatened (NT) & Least Concern (LC).**

7.2 Flagship Endemic Species of India

- **Lion-Tailed Macaque (*Macaca silenus*):** Arboreal primate endemic exclusively to the tropical evergreen rainforest canopy of the Western Ghats.
- **Nilgiri Tahr (*Nilgiritragus hylocrius*):** Mountain ungulate endemic to the high-altitude shola-grassland ecosystem of the Western Ghats (Eravikulam National Park).
- **Sangai (Brow-Antlered Deer):** Endemic exclusively to Keibul Lamjao National Park (the world's only floating national park on Loktak Lake, Manipur).

Quick Recall:

IUCN Red List: Extinct → Critically Endangered → Endangered → Vulnerable. Endemic species: Lion-tailed Macaque (Western Ghats), Sangai Deer (floating Loktak lake, Manipur).

8. In-Situ vs. Ex-Situ Conservation Strategies

Conservation Paradigm	Definition & Core Philosophy	Implementation Modalities & Legal Protection
In-Situ Conservation (On-Site Preservation)	Conserving genetic resources, wild species, and ecosystems directly within their natural native habitats where evolutionary processes continue.	• National Parks (IUCN Cat II): Strict protection. Zero human settlement, livestock grazing, or private exploitation permitted (e.g. Kaziranga, Corbett, Bandipur). • Wildlife Sanctuaries (IUCN Cat IV): Protection focused on particular wildlife species; limited traditional forestry and grazing permitted under permission. • Biosphere Reserves (UNESCO MAB): Large landscapes consisting of 3 zones: <i>Core Zone</i> (pristine/untouched), <i>Buffer Zone</i> (research/education), and <i>Transition Zone</i> (sustainable community living). 18 in India (Nilgiri, Nanda Devi). • Sacred Groves: Traditional community-protected forest patches. • Flagship Projects: Project Tiger (1973), Project Elephant (1992).
Ex-Situ Conservation (Off-Site Preservation)	Conserving endangered species outside their natural habitats in artificial, human-managed captive environments to safeguard genetic stock against imminent extinction.	• Zoological Parks & Botanical Gardens: Captive breeding and propagation. • Seed Banks & Gene Banks: Long-term storage of desiccated plant seeds at -20°C (e.g. National Bureau of Plant Genetic Resources – NBPGR, New Delhi). • Cryopreservation: Preserving gametes, embryos, and vegetative cells in liquid nitrogen at -196°C for indefinite periods. • In-Vitro Tissue Culture: Clonal micropropagation of endangered flora.

Quick Recall:

In-situ = on-site (National Parks [no grazing], Sanctuaries, Biosphere Reserves [Core, Buffer, Transition]). Ex-situ = off-site (Botanical gardens, Gene banks, Cryopreservation at -196°C).

9. Solved Diversity Index Calculations

Solved Problem 3.1: Calculating Simpson's Diversity Index:

Problem Statement: An ecologist surveys a 1-hectare sample plot in the Western Ghats and records the following tree species counts:

- Species A (Teak): $n_1 = 40$
- Species B (Rosewood): $n_2 = 25$
- Species C (Sandalwood): $n_3 = 15$
- Species D (Ebony): $n_4 = 12$
- Species E (Ironwood): $n_5 = 8$

Calculate: 1. Simpson's Index of Dominance (D); 2. Simpson's Index of Diversity ($1 - D$); 3. Simpson's Reciprocal Index ($\frac{1}{D}$).

Step 1: Compute Total Number of Individuals (N):

$$N = 40 + 25 + 15 + 12 + 8 = 100$$

$$N(N - 1) = 100 \times 99 = 9,900$$

Step 2: Calculate $n_i(n_i - 1)$ for Each Species:

- Species A: $40 \times 39 = 1,560$
- Species B: $25 \times 24 = 600$
- Species C: $15 \times 14 = 210$
- Species D: $12 \times 11 = 132$
- Species E: $8 \times 7 = 56$

$$\sum n_i(n_i - 1) = 1,560 + 600 + 210 + 132 + 56 = 2,558$$

Step 3: Compute Simpson's Index (D):

$$D = \frac{\sum n_i(n_i - 1)}{N(N - 1)} = \frac{2,558}{9,900} \approx 0.2584$$

Step 4: Compute Simpson's Index of Diversity ($1 - D$) and Reciprocal Index ($\frac{1}{D}$):

$$\text{Simpson's Diversity } (1 - D) = 1 - 0.2584 = 0.7416$$

$$\text{Reciprocal Index } \left(\frac{1}{D}\right) = \frac{1}{0.2584} \approx 3.870$$

$$D = 0.2584, \quad \text{Simpson's Diversity } (1 - D) = 0.7416, \quad \frac{1}{D} = 3.870$$

Ecological Interpretation: There is a 74.16% probability that two randomly selected trees from this forest plot belong to different species, indicating high community diversity.

Species	Count (n_i)	Proportion (p_i)	$\ln(p_i)$	$p_i \ln(p_i)$
A	40	0.40	-0.9163	-0.3665
B	25	0.25	-1.3863	-0.3466
C	15	0.15	-1.8971	-0.2846
D	12	0.12	-2.1203	-0.2544
E	8	0.08	-2.5257	-0.2021
Sum	100	1.00	-	-1.4541

Species Tag	Abundance (n_i)	Proportion (p_i)	$n_i(n_i - 1)$	Contribution
Species A	40	0.4000	1560	15.8%
Species B	25	0.2500	600	6.1%
Species C	15	0.1500	210	2.1%
Species D	12	0.1200	132	1.3%
Species E	8	0.0800	56	0.6%

10. Quick Revision Sheet

Concept	Key Criteria / Formula	Must-Know Exam Facts
Levels of Biodiversity	Genetic, Species (richness + evenness), Ecosystem (α, β, γ).	India has > 50,000 strains of rice.
Biodiversity Hotspots	$\geq 1,500$ endemic vascular plants AND lost $\geq 70\%$ habitat.	4 in India: Western Ghats, Eastern Himalayas, Indo-Burma, Sundaland.
The Evil Quartet	Habitat destruction (primary), Over-exploitation, Alien invasives, Co-extinctions.	Invasive species: <i>Lantana</i> , <i>Parthenium</i> , <i>Eichhornia</i> .
In-Situ vs Ex-Situ	In-situ = on-site (National Parks); Ex-situ = off-site (Cryopreservation).	National parks allow ZERO human settlement or grazing.
Simpson's Diversity	$D = \sum \frac{n_i(n_i-1)}{N(N-1)}$, Diversity = $1 - D$.	Higher $1 - D$ indicates greater community diversity.

Concept	Key Criteria / Formula	Must-Know Exam Facts
Shannon-Wiener Index	$H' = - \sum p_i \ln(p_i), \quad J' = \frac{H'}{\ln(S)}$	Captures both species richness and equitability.

11. Authentic Exam Archive & Model Solutions

The following conceptual questions and verified diversity metric calculations cover key university examination patterns for HS 510: Environmental Studies:

2-Mark Conceptual Questions

Practice Problem:

2 MARKS 1. Define the term 'Endemic Species' with an authentic example from India.

Solution:

Answer:

Endemic Species are plants or animals whose natural wild geographic distribution is strictly confined to a specific geographic region or habitat, and found nowhere else globally.

Authentic Indian Example: The Lion-tailed Macaque (*Macaca silenus*) and the Nilgiri Tahr (*Nilgiritragus hylocrius*), both endemic exclusively to the tropical evergreen rainforests of the Western Ghats.

Practice Problem:

2 MARKS 2. State the two essential criteria proposed by Norman Myers to designate a global Biodiversity Hotspot.

Solution:

Answer:

- 1. High Endemism Criterion:** The region must contain at least **1,500 species of endemic vascular plants** ($> 0.5\%$ of the world's total plant species).
- 2. High Threat / Habitat Loss Criterion:** The region must have lost at least **70% of its original native primary vegetation** (i.e., retaining $\leq 30\%$ of its pristine natural habitat).

 Practice Problem:

2 MARKS 3. What are the four global Biodiversity Hotspots that cover regions of India?

Solution:

Answer:

1. **Western Ghats and Sri Lanka** (Southwestern India).
2. **Eastern Himalayas** (Covering Arunachal Pradesh, Sikkim, and adjacent sub-Himalayan tracts).
3. **Indo-Burma** (Covering North-Eastern states including Manipur, Mizoram, Nagaland).
4. **Sundaland** (Covering the Nicobar Islands group).

 Practice Problem:

2 MARKS 4. Differentiate between Alpha, Beta, and Gamma diversity in landscape ecology.

Solution:

Answer:

- **Alpha (α) Diversity:** Species diversity within a particular local community or single habitat.
- **Beta (β) Diversity:** The rate of species turnover between two distinct adjacent communities or habitats along an environmental gradient.
- **Gamma (γ) Diversity:** The overall total species richness across an entire landscape, biome, or geographic region ($\gamma = \alpha + \beta$).

5-Mark & 10-Mark Model Questions

Dimension	In-Situ Conservation (On-Site)	Ex-Situ Conservation (Off-Site)
Definition	Conservation of species within their natural native habitats where evolutionary processes continue.	Conservation of endangered organisms in artificial, human-controlled environments outside native habitats.
Examples	National Parks (Bandipur, Kaziranga), Wildlife Sanctuaries, Biosphere Reserves (Nilgiri), Sacred Groves.	Zoological Parks, Botanical Gardens, Seed Banks (NBPGR, New Delhi), Cryopreservation facilities.
Human Intervention	Minimal; National Parks prohibit all human settlement, grazing, and resource extraction.	Intensive; organisms depend upon artificial feeding, veterinary care, and controlled temperature regimes.

Dimension	In-Situ Conservation (On-Site)	Ex-Situ Conservation (Off-Site)
Evolutionary Value	High; populations continue natural adaptation, predator-prey dynamics, and speciation.	Low; natural selection halted; vulnerability to genetic bottlenecks and inbreeding depression.
Scope & Cost	Protects entire ecosystems, thousands of species, and trophic pyramids cost-effectively.	Focuses on select high-profile endangered taxa; requires high capital and maintenance expenditures.

Practice Problem:

8 MARKS 6. Explain the four major drivers of biodiversity loss under Edward O. Wilson's 'Evil Quartet'.

Solution:

- Habitat Loss & Fragmentation (Primary Cause):** Clearing pristine tropical rainforests for commercial monocultures (palm oil, soy) and infrastructure fractures contiguous habitats into tiny isolated forest islands, causing edge effects and local population extinctions.
- Over-Exploitation:** Excessive harvesting exceeding biological reproduction rates. Driven by greed rather than need (e.g. historical extinction of the Passenger Pigeon, Steller's Sea Cow, and poaching of One-horned Rhinos for horn keratin).
- Invasion of Alien Species:** Accidental or intentional introduction of non-native exotic species that aggressively outcompete native taxa. Examples: *Parthenium hysterophorus* (congress grass), *Lantana camara*, and *Eichhornia crassipes* (water hyacinth / "Terror of Bengal") choking aquatic ecosystems.
- Co-Extinctions:** When an obligate mutualistic relationship exists, the extinction of a host species unavoidably triggers the co-extinction of its obligate pollinator or parasite (e.g. co-evolved fig-wasp mutualisms).

Verified Numerical & Computational Practice Problems

PRACTICE 1 · SIMPSON'S INDEX OF DIVERSITY

A sample of 100 trees from a forest plot contains 5 species with abundances: 40, 25, 15, 12, and 8. Calculate Simpson's Index (D), Simpson's Index of Diversity ($1 - D$), and Reciprocal Index ($1/D$).

 Practice Problem:

Show Step-by-Step Solution

Solution:

1. Total individuals: $N = 40 + 25 + 15 + 12 + 8 = 100$.
2. Denominator: $N(N - 1) = 100 \times 99 = 9,900$.
3. Numerator: $\sum n_i(n_i - 1) = (40 \times 39) + (25 \times 24) + (15 \times 14) + (12 \times 11) + (8 \times 7) = 1560 + 600 + 210 + 132 + 56 = 2,558$.
4. Simpson's Dominance: $D = \frac{2,558}{9,900} \approx \mathbf{0.2584}$.
5. Simpson's Diversity: $1 - D = 1 - 0.2584 = \mathbf{0.7416}$.
6. Reciprocal Index: $\frac{1}{D} = \frac{1}{0.2584} \approx \mathbf{3.870}$.

Verification: Verified by automated unit test in
audit/verify/evs/verify_evs_u3.py.

PRACTICE 2 · SHANNON-WIENER DIVERSITY & EVENNESS

For the identical tree community ($N = 100$, $S = 5$, counts: 40, 25, 15, 12, 8), calculate the Shannon-Wiener Diversity Index (H') and Pielou's Evenness (J').

 Practice Problem:

Show Step-by-Step Solution

Solution:

Proportions: $p_i = [0.40, 0.25, 0.15, 0.12, 0.08]$

- $0.40 \ln(0.40) = 0.40 \text{ times } (-0.9163) = -0.3665$
- $0.25 \ln(0.25) = 0.25 \text{ times } (-1.3863) = -0.3466$
- $0.15 \ln(0.15) = 0.15 \text{ times } (-1.8971) = -0.2846$
- $0.12 \ln(0.12) = 0.12 \text{ times } (-2.1203) = -0.2544$
- $0.08 \ln(0.08) = 0.08 \text{ times } (-2.5257) = -0.2021$
- Sum = $-1.4541 \implies H' = 1.4541$.

Maximum possible diversity: $H_{\text{extmax}} = \ln(S) = \ln(5) = 1.6094$.

Species Evenness: $J' = \frac{H'}{H_{\text{extmax}}} = \frac{1.4541}{1.6094} \approx \mathbf{0.9035}$ (90.35% equitable distribution).

Verification: Verified by automated unit test in
audit/verify/evs/verify_evs_u3.py.

PRACTICE 3 · THEORETICAL MAXIMUM DIVERSITY

Under what ecological condition does Shannon-Wiener Diversity (H') reach its absolute theoretical maximum H_{extmax} ?

 Practice Problem:

Show Step-by-Step Solution

Solution:

Shannon-Wiener diversity reaches $H_{\text{extmax}} = \ln(S)$ when all S species have strictly equal abundances (i.e. $p_i = \frac{1}{S}$ for all i). Under this condition, Pielou's evenness $J' = \frac{\ln(S)}{\ln(S)} = \mathbf{1.00}$ (100% perfect community equitability).

PRACTICE 4 · CRYOPRESERVATION MECHANISMS

Explain the thermodynamic principle and storage temperature utilized in cryopreservation for ex-situ biodiversity conservation.

 Practice Problem:

Show Step-by-Step Solution

Solution:

Cryopreservation preserves plant germplasm (pollen, seeds, meristems) and animal gametes/embryos in **liquid nitrogen at -196°extC (77extK)**.

Principle: At -196°extC , all chemical kinetic activity and cellular metabolic respiration cease completely. Biological tissues remain in metabolic suspended animation for decades to centuries without genetic degradation or senescence.

PRACTICE 5 · BIOGEOGRAPHIC ZONES OF INDIA

Enumerate the 10 Biogeographic Zones of India designated by the Wildlife Institute of India (WII).

 Practice Problem:

Show Step-by-Step Solution

Solution:

1. 1. Trans-Himalayan Zone (Ladakh, Lahaul-Spiti)
2. 2. Himalayan Zone (Northwest, West, Central, and East Himalayas)
3. 3. Desert Zone (Thar and Kutch)
4. 4. Semi-Arid Zone (Punjab plains to Gujarat)
5. 5. Western Ghats (Malabar plains and Sahyadri ranges)
6. 6. Deccan Peninsula (Central highlands, Chhota Nagpur, Eastern ghats)
7. 7. Gangetic Plain (Upper and Lower Gangetic alluvial plains)
8. 8. Coastal Zone (West and East coasts, Lakshadweep)
9. 9. North-East Zone (Brahmaputra valley and Assam hills)
10. 10. Islands (Andaman and Nicobar archipelago)

PRACTICE 6 · INVASIVE ALIEN SPECIES (IAS) DISRUPTION

Describe how **Eichhornia crassipes** (Water Hyacinth) alters aquatic ecology and drives local extinctions.

 Practice Problem:

Show Step-by-Step Solution

Solution:

Eichhornia crassipes (Water Hyacinth), introduced as an ornamental aquatic plant, forms dense impenetrable floating vegetative mats over freshwater lakes and rivers:

- **Sunlight Blockade:** Blocks solar penetration, killing submerged photosynthesizing aquatic macrophytes and phytoplankton.
- **Dissolved Oxygen (DO) Depletion:** High biomass decay exerts massive Biochemical Oxygen Demand (BOD), causing severe hypoxia/anoxia that asphyxiates native fish fauna.
- **Ecosystem Succession Acceleration:** Accelerated sediment buildup converts open water bodies into shallow marshes prematurely.

12. Prescribed References

Course Reference Literature:

1. Erach Bharucha, *Textbook of Environmental Studies for Undergraduate Courses*, Universities Press / UGC India. (Chapter 4: Biodiversity).
 2. Benny Joseph, *Environmental Studies*, Tata McGraw-Hill Education.
 3. R. Rajagopalan, *Environmental Studies: From Crisis to Cure*, Oxford University Press.
- AI-assisted study notes written from the official courseware. Verify against your faculty's presentation slides before the exam.